

Eskom restructuring poses powerful dilemmas

Anton Eberhard and Fiona Thompson

BUSINESS DAY 18 October 2001

A study by investment strategy group *BusinessMap* has brought into sharp focus the range of options available to government in restructuring the electricity supply industry. Cheap, reliable power is one of South Africa's key competitive advantages in the global economy, but no matter what, wholesale electricity prices will rise.

Eskom can partly take credit for present low prices resulting from productivity improvements. But massive over-investment by Eskom in electricity generation plant in the 1970s and 1980s is the main reason. Little new investment has been required in recent years and so financing costs have fallen sharply. Eskom's debt to equity ratio has declined from 2.93 in 1986 to 0.63 in 2000.

This apartheid-era inheritance is running out. South Africa will soon run out of surplus generating capacity (between 2007 and 2011) and because of the long lead times in generation investments, decisions on new investments will need to be made soon. Generation capacity will have to more or less double in the next 25 years to meet estimated future demand. This will require an investment in new electricity generation plant of around R100bn over the period.

Electricity prices are below the level required to support new investment. Prices will rise no matter how the electricity industry is organised in the future. Two immediate challenges arise: first, how do we manage prices to a level which will support new investment without causing price shocks when capacity runs out and, second, how do we create an industry structure which will promote new investments most efficiently and at the least cost?

The introduction of competitive markets should improve efficiency in allocating capital and in operating plant and hence lead to low costs. Low costs will lead to low prices if these benefits are passed to consumers. A revolution in the electricity industry internationally has demonstrated that competition is possible in the generation and supply of electricity. Wholesale and retail electricity prices, when deregulated, have fallen in many countries; recent examples are Germany and New Zealand.

Yet the *BusinessMap* study shows that a competitive market structure in South Africa might actually result in *higher* prices in the medium term than if the present regulated and integrated Eskom monopoly continues.

This is because the calculation of regulated prices is based on historical asset values and average costs in the industry. Much of the existing asset base is old and therefore heavily depreciated. Original asset values have also been eroded by inflation. The low asset values of older plant serve to keep average costs, and hence average prices, low. Once new investment takes place and the mix of assets changes from old to new assets, so average costs will tend to rise.

In a competitive market, prices are unregulated and essentially based on the interaction of supply and demand. Once surplus generating capacity has been eroded by increases in demand, prices will move quickly to around the cost for new entrants (i.e. the long-run marginal cost) and this attracts new investment.

In regulated markets, new investment in plant is planned, based on demand forecasts and *consumers* therefore bear the risks of mistakes. A key element of competitive markets is that since prices trigger new investments, *investors* bear the risks of this investment. This is the feature of competition that should lead to efficient investment decision-making.

The modelling in the study assumes international benchmark prices for new generation investments and that the operational efficiency gains in a competitive market are likely to be modest, given our low fuel costs and high productivity levels.

Two trends could bring regulated prices closer to competitive market prices. Firstly, the pricing methodology now used by the National Electricity Regulator could change to incorporate higher rates of return on an optimised replacement value of assets. In this case, regulated prices might approach those necessary to support new investment. Secondly, in the long run, unregulated competitive markets should lead to lower costs than regulated ones. Competitive pressures should keep operating costs low, and market risks should encourage investment at optimal levels. Ultimately this should lead to lower prices, but not necessarily during a “transition” period as older capacity is phased out and new capacity is phased in.

Estimated future electricity prices affect valuations of assets. The *BusinessMap* study indicates some remarkable results in the potential valuation of Eskom generating assets. Values vary between R24bn and R140bn, depending on market structure and the level of competition.

In unregulated markets prices will rise to a level until they provide an adequate return to incentivise new investment. At these prices, older fully depreciated assets still in operation will earn extremely high returns and consequently, towards the end of their operating lives will become very profitable. This in effect represents a transfer from consumers (paying the market price) to the owners of this low-cost, older plant.

Mechanisms to avoid this transfer have been quite common in other countries that have restructured their electricity industries. They typically take the form of contracts between existing plant and power purchasers, vested at the time of reform and fixing prices based on a formula. Thus South Africa might begin to move to an industry structure where market prices begin to signal the marginal cost of new generating plant and therefore promote efficient additional investment but in the transition period vested contracts can protect consumers from sharp price increases.

The *BusinessMap* study also indicates the scope for restructuring to meet other government objectives such as black economic empowerment, attracting foreign direct investment and realising privatisation receipts to reduce public debt. Lower interest payments on this debt could free additional fiscal resources for social investment in electrification as well as for restructuring local government finances to replace the municipal surcharge on electricity sales. In this way the upward pressure on electricity prices for consumers could be reduced.

The restructuring of Eskom presents government with opportunities to meet a wide spectrum of economic and social policy goals. Government will have to manage the restructuring process carefully though to maintain competitive electricity prices and at the same time lay the groundwork for efficient new investment in the industry.

Anton Eberhard is a professor at the Graduate School of Business at the University of Cape Town. Dr Fiona Thompson is a senior energy consultant at BusinessMap, Johannesburg.